Rajshahi University of Engineering & Technology (RUET)
Department of Computer Science and Engineering (CSE)
CSE 2202: Algorithm Analysis and Design Lab
Lab Final Examination Time: 1 Hour Total Marks: 20

Problem: Interstellar Message & Path Analysis

You are an analyst at the RUET Space Exploration Lab. You have received two garbled message fragments, Msg_A and Msg_B, from a deep-space probe. You also have a map of a newly discovered wormhole network, which your probe must navigate.

Your goal is to solve two critical problems using **Dynamic Programming**:

1. **Similarity Analysis:** Find the largest common message "core" hidden within the two garbled fragments.
2. **Route Optimization:** Find the "cheapest" path for the probe to travel from a START base to all other bases in the wormhole network.

Task 1: Message Core Similarity [10 Marks]

The two message fragments are:

Msg_A = "AGTCAATTGC"
Msg_B = "GATTCAG"

The "message core" is the **longest possible sequence of characters** that appears as a subsequence in *both* Msg_A and Msg_B.

a) [7 Marks] Implement a dynamic programming solution to find the length of the longest common message core.

- Output 1: Print the entire filled 2D DP table.
- Output 2: Print the final length of the longest common core.

b) [3 Marks] Reconstruct and print the actual common core.

- Output 3: Print one valid sequence for the longest common core.

Task 2: Wormhole Route Optimization [10 Marks]

The probe must navigate a complex wormhole network. The map is a **directed, weighted graph** where nodes are space bases and edges are wormholes. The edge weights represent the "energy cost" to traverse them.

Critically, some wormholes have **exotic matter** that provides energy to the probe, resulting in a **negative travel cost**.

Graph Data:

Source Base: Node 0
Bases (Nodes): 0, 1, 2, 3, 4, 5
Wormholes (Edges) as (from, to, cost):

- (0, 1, 6)
- (0, 2, 5)
- (1, 3, 3)
- (1, 4, -3) *(Energy-gaining path)*
- (2, 1, -2) *(Energy-gaining path)*
- (3, 2, 4)
- (3, 5, 3)
- (4, 5, 2)
- (5, 1, -1) *(Energy-gaining path)*
- (5, 3, -4) *(Energy-gaining path)*

Your task is to find the minimum energy cost from the Source Base (0) to all other bases.

a) **[7 Marks]** Implement a dynamic programming solution based on iterative relaxation.

➢ Output 4: If no "infinite energy loops" exist, print the minimum energy cost to reach each Base (0, 1, 2, 3, 4, and 5) from the Source Base 0.

b) **[3 Marks]** Your algorithm must also be able to detect if the network contains an "infinite energy loop" (a cycle of wormholes where the total cost is negative, allowing the probe to gain infinite energy).

➢ Output 5: If such a loop is detected, print "NEGATIVE CYCLE DETECTED". (If this happens, the costs from Output 4 may be invalid, but your detection is the key.)